

TrES-5b

R-1554

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130406 · created 2026-08-01T20:18:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456388.9243 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.052
Transit depth (Rp/Rs)^2	1.57 +/- 1.3 [%]
Orbital Inclination (inc)	82.45 +/- 2.65
Airmass coefficient 1 (a1)	1.157 +/- 0.087
Airmass coefficient 2 (a2)	-0.07 +/- 0.017
Scatter in the residuals of the lightcurve fit is	7.72 %
Best Comparison Star	#2 - [332, 253]
Optimal Aperture	2.16
Optimal Annulus	4.83
Transit Duration (day)	0.053 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.052 vs 0.142 (deviation 0.21σ)
Duration fit vs published	0.053 d vs 0.0758 d (deviation 0.54σ)
Mid-time O-C	-3.4 min
Depth SNR	1.5
Residual scatter	7.72 %

Light curve

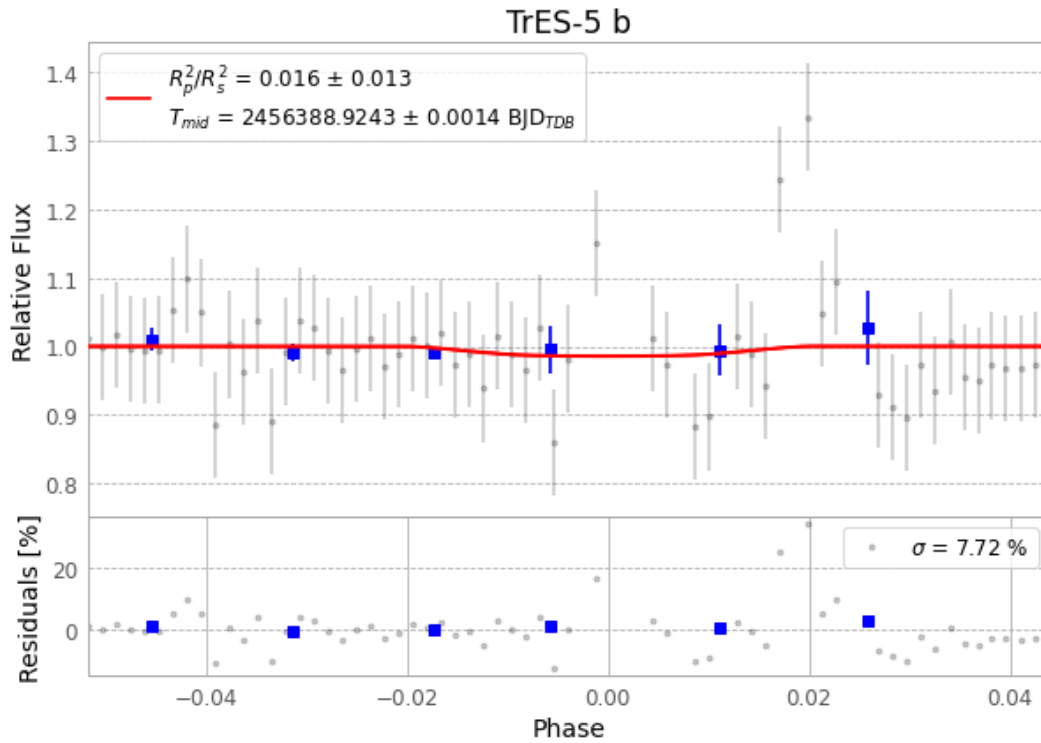


Figure 1 — FinalLightCurve_TrES-5 b_2013-04-06.png (SHA-256 3d94c4932ae816ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 117], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 043f9be349a331fc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-5130406082112.FITS → TRES-5130406114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130406.log (1e503f83f0cd...), FinalLightCurve_TrES-5 b_2013-04-06.png (3d94c4932ae8...), FinalLightCurve_TrES-5 b_2013-04-06.csv (f01ead320852...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:18Z.